

Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-15: Wavelet and Multiresolution Processing

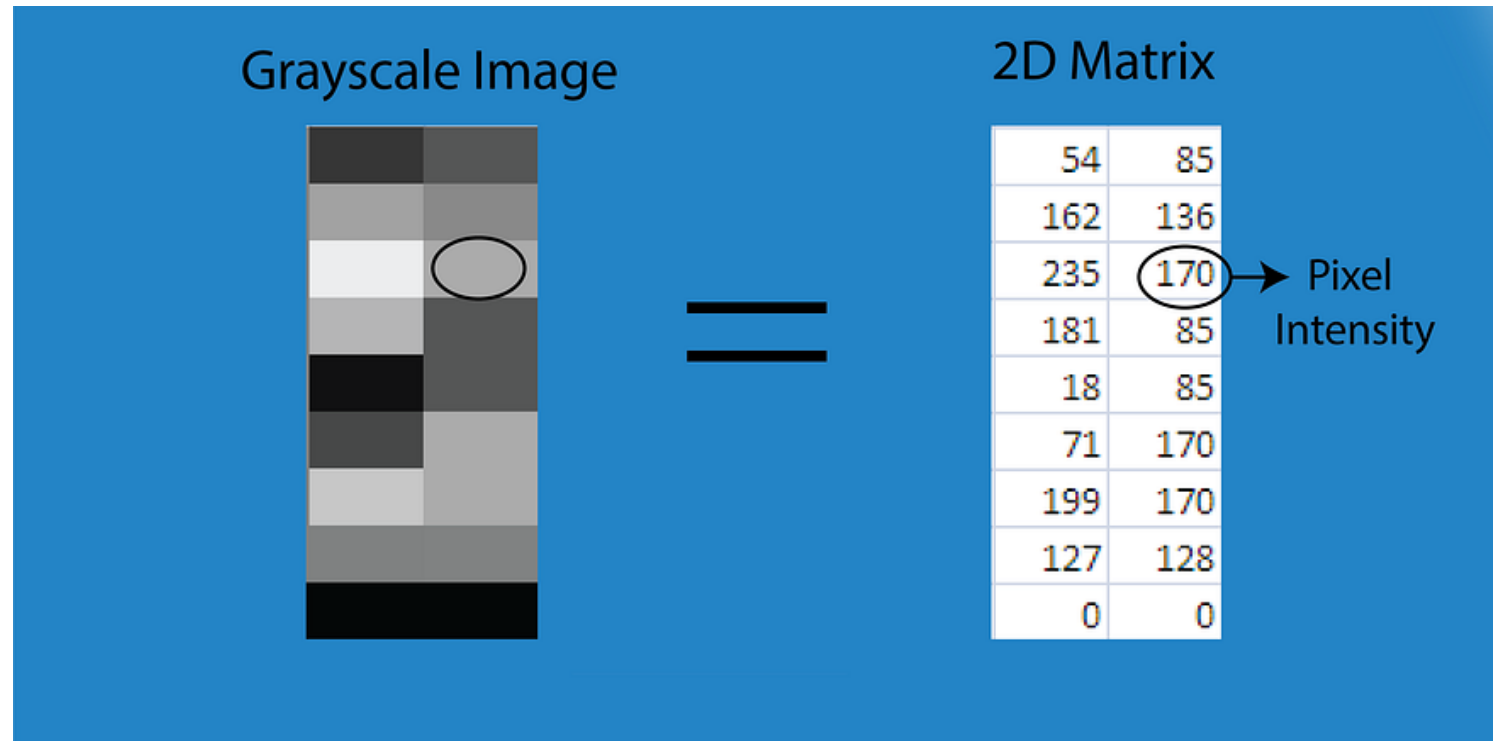
Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-7)

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Spatial Domain

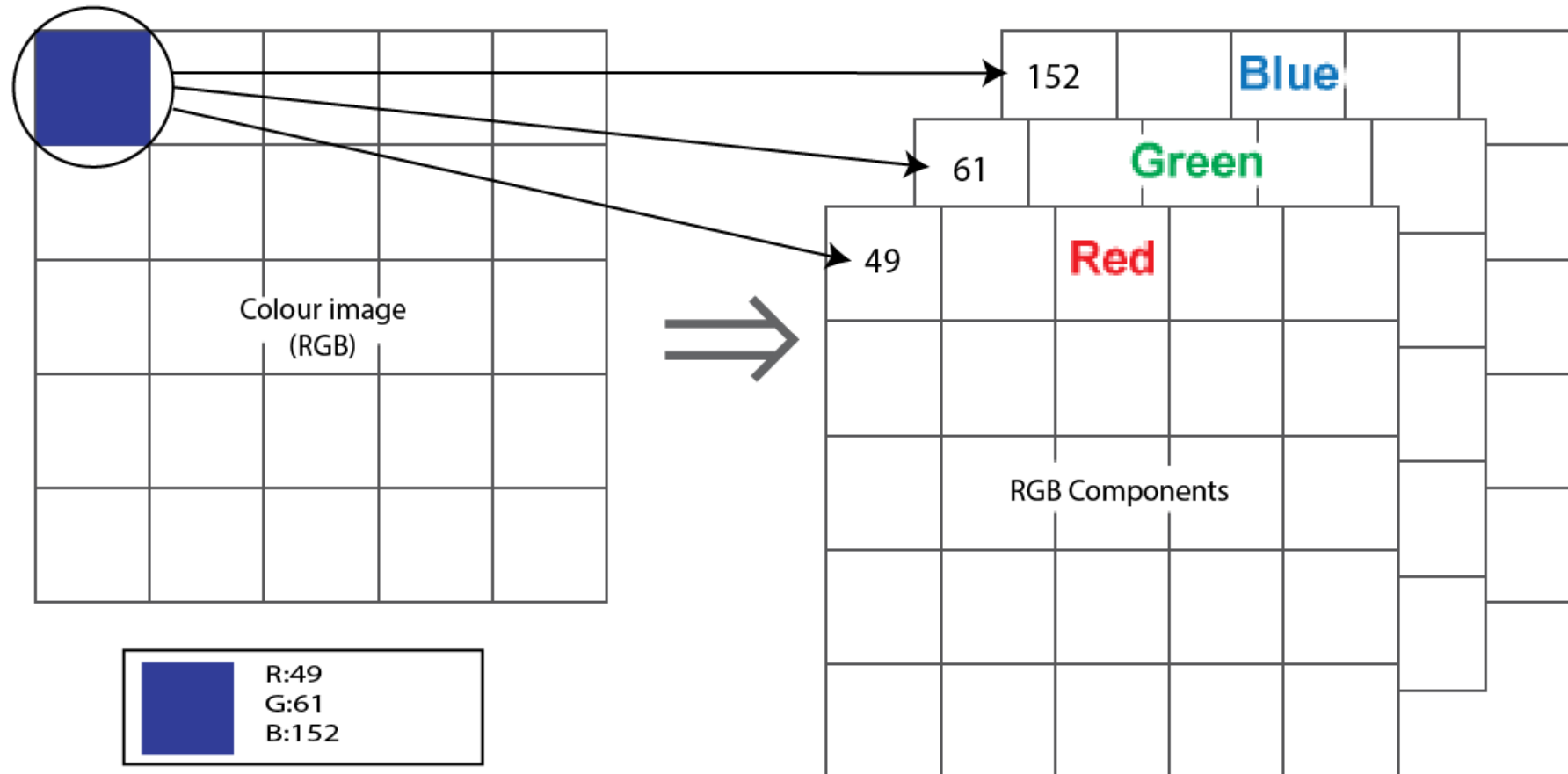
An image can be represented in the form of a 2D matrix where each element of the matrix represents pixel intensity.

This state of 2D matrices that depict the intensity distribution of an image is called Spatial Domain.



For the RGB image, the spatial domain is represented as a **3D vector of 2D matrices**.

Each 2D matrix contains the intensities for a single color as shown below-



Mathematical Expression

Each pixel intensity is represented as $I(x,y)$ where x,y is the co-ordinate of the pixel in the 2D matrix.

Different operations are carried out in this value.

For example- operation T (say, addition of 5 to all the pixel) is carried out in $I(x,y)$ which means that each pixel value is increased by 5. This can be written as-

$$I'(x,y) = T[I(x,y)]$$

where, $I'(x,y)$ is the new intensity after adding 5 to $I(x,y)$.

Frequency Domain

In frequency-domain methods are based on Fourier Transform of an image. Roughly, the term frequency in an image tells about the rate of change of pixel values.

In RGB image, the frequency simply refers to the rate of change of color components in an image.

Areas of high frequencies experience rapid color changes, whereas parts that change gradually contain low frequencies.

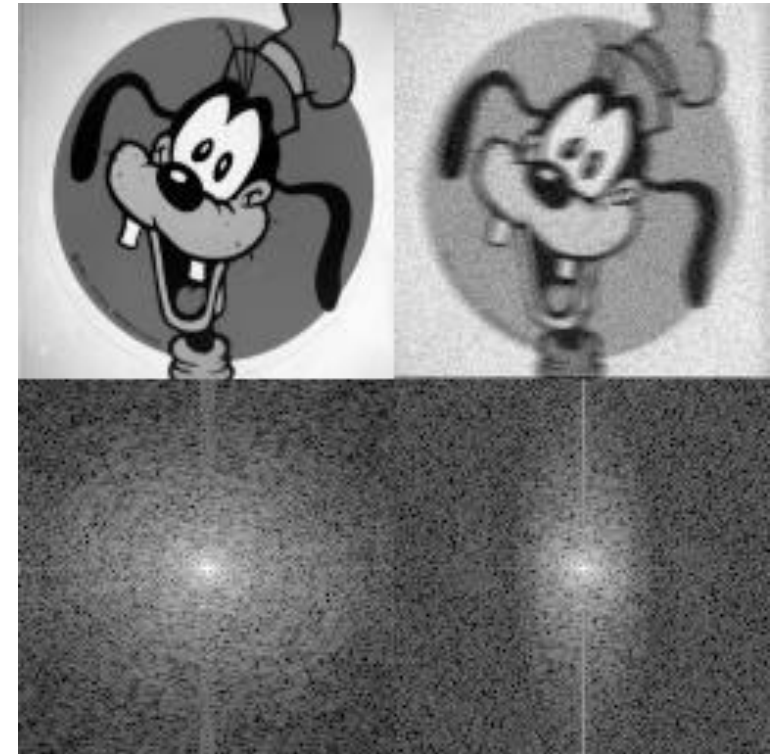
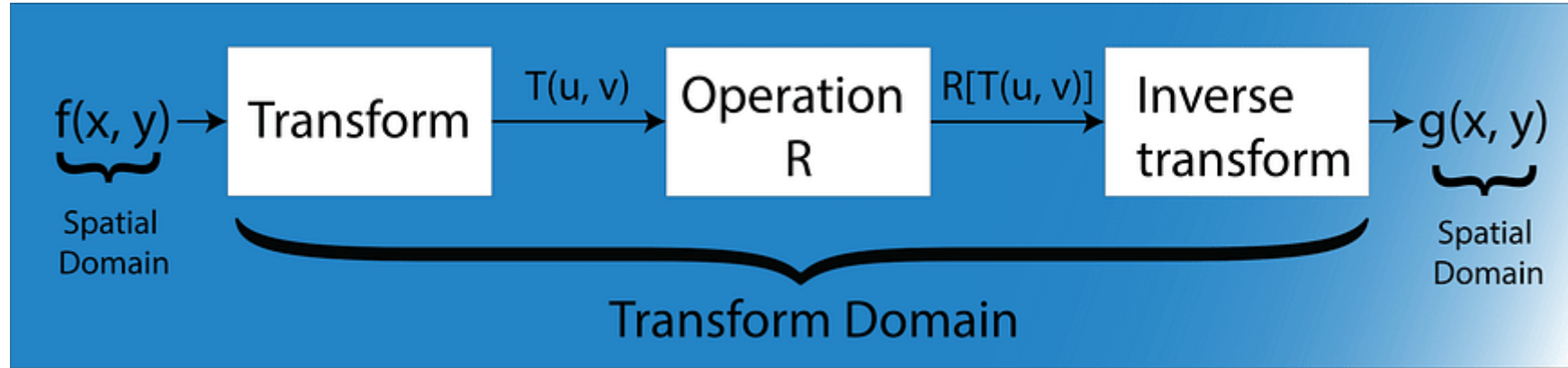


Image Transformation

Image Transformation mainly follows three steps-



Step-1. Transform the image.

Step-2. Carry the task(s) in the transformed domain.

Step-3. Apply inverse transform to return to the spatial domain.

Unlike spatial domain, we cannot directly operate on the values. The image is first transformed to its frequency distribution before processing it. These frequency components are divided into two major components.

*** High Frequency components that correspond to edges in an image and low frequency component that correspond to smooth regions.

The output of this process isn't an image, rather a transformation. To reconstruct the image to its ideal form, we need to apply inverse transformation on the processed output.

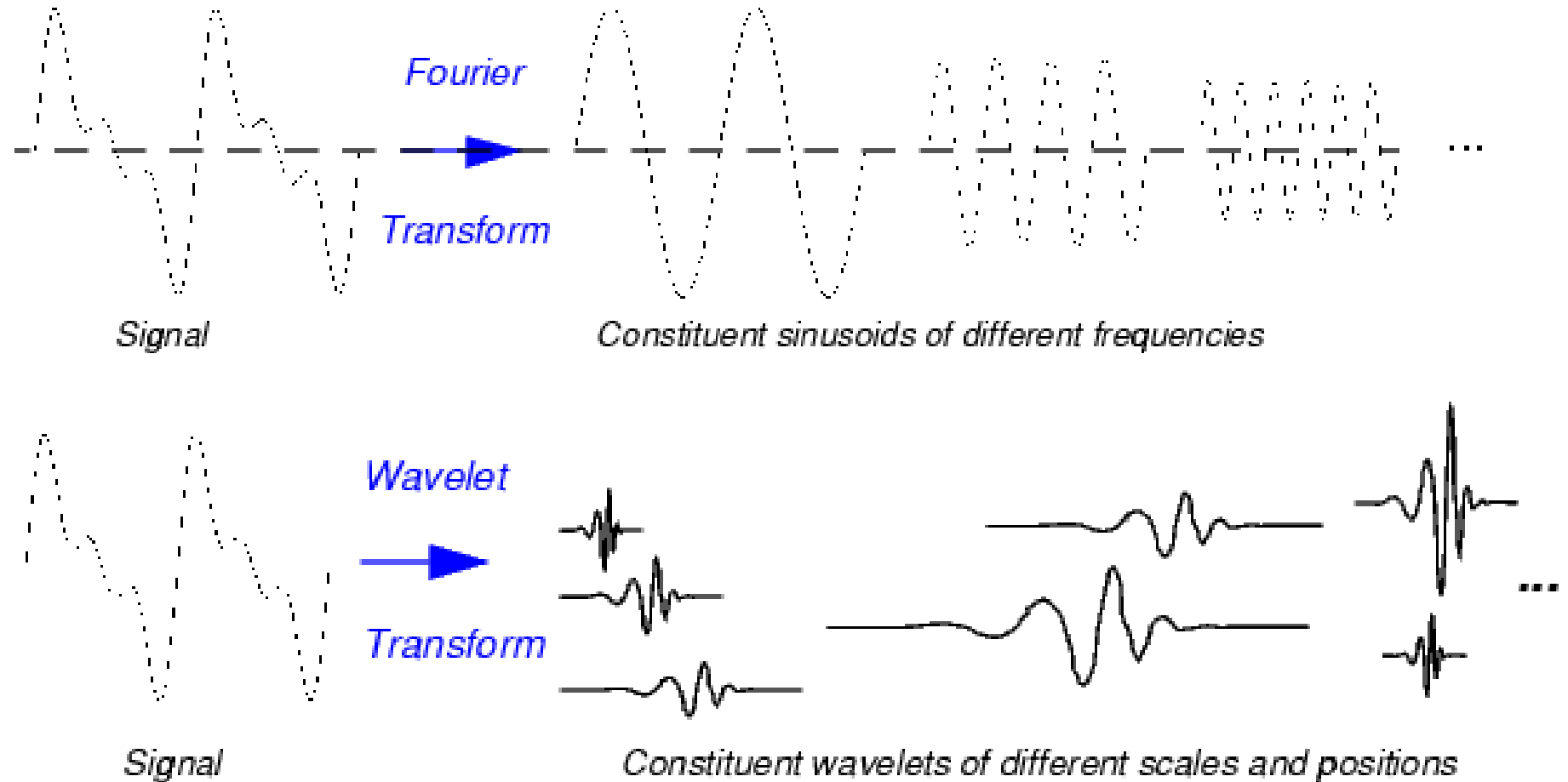
There are several mathematical transforms under frequency domain such as the

- ❑ **Fourier transform**
- ❑ Laplace transform
- ❑ Z transform
- ❑ **Wavelet Transform**

The wavelet transformation technique that is generally used for image analysis and data compression.

1. Wavelet

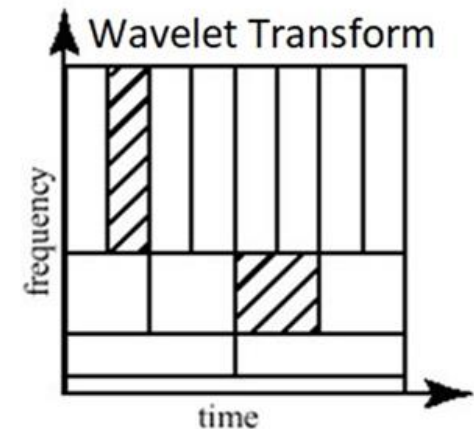
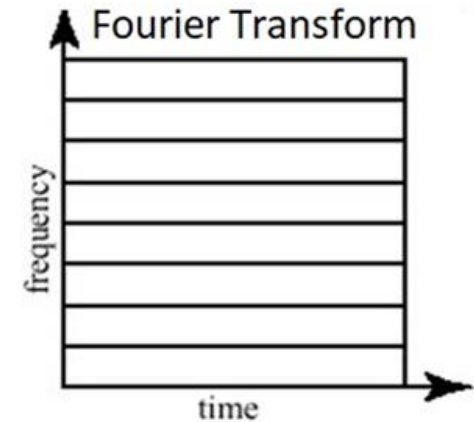
Unlike the Fourier transform, whose basis functions are sinusoids, wavelet transforms are based on small waves, called wavelets, of varying frequency and limited duration.



2. Wavelet Transform

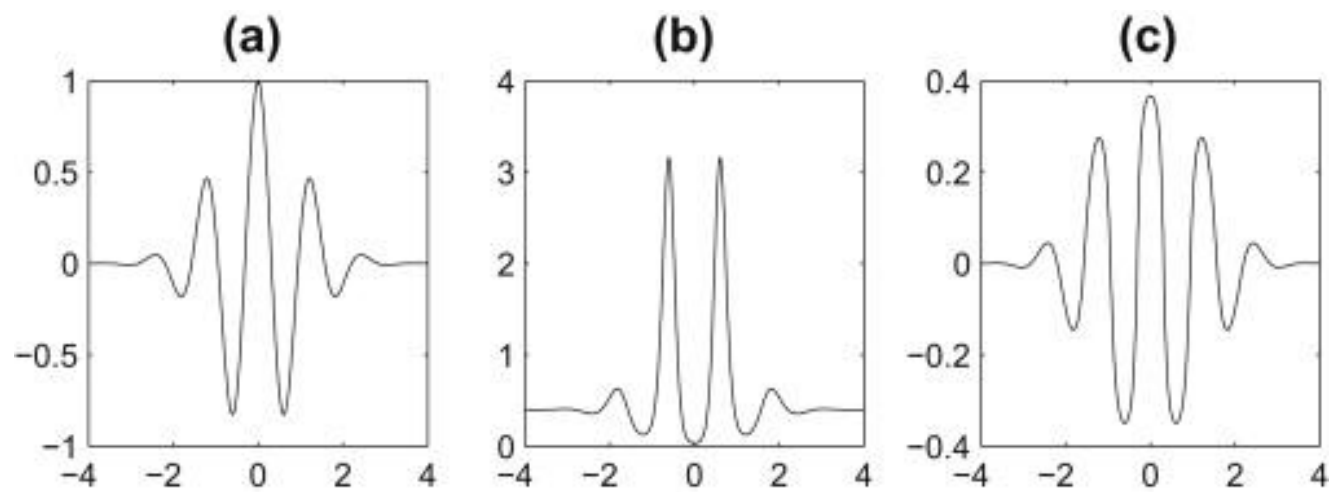
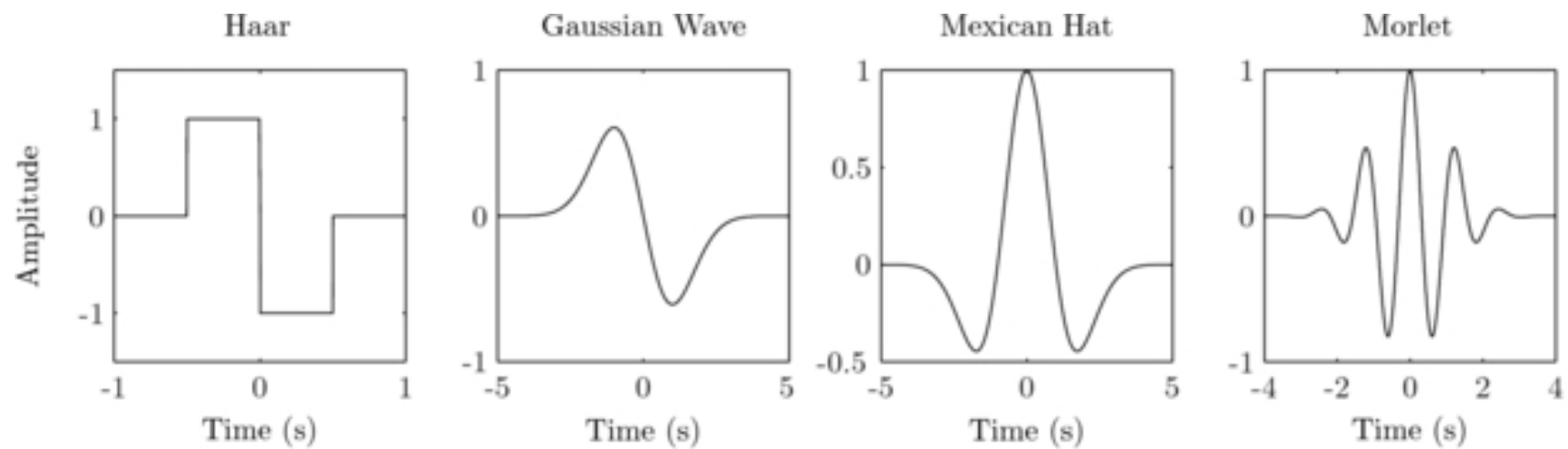
Wavelet Transform (WT) is a mathematical technique that decomposes a signal (or image) into different frequency components while maintaining spatial (or time) information.

Unlike traditional Fourier Transform, which only provides frequency information, WT provides both frequency and spatial (or time) resolution.



This method starts with a *mother wavelet such as Haar, Morlet, Daubechies*, etc. The signal is then essentially *translated into scaled and shifted* versions of mother wavelet.

Continuous Wavelet Transform (CWT)	Discrete Wavelet Transform (DWT)
Mexican Hat Wavelet	Haar
Morlet Wavelet	Daubechies (order 1, 2, 4, and 8)
Shannon Wavelets	Symlets (order 1, 2, 4, and 8)
Frequency B-Spline Wavelets	Coeiflets (order 1 and 2)



Fourier Transform

The basis functions are sinusoids.

Converts signals into a continuous series of sine and cosine functions with constant frequency and amplitude.

Provides only frequency information.

Temporal (time) information is lost during transformation.

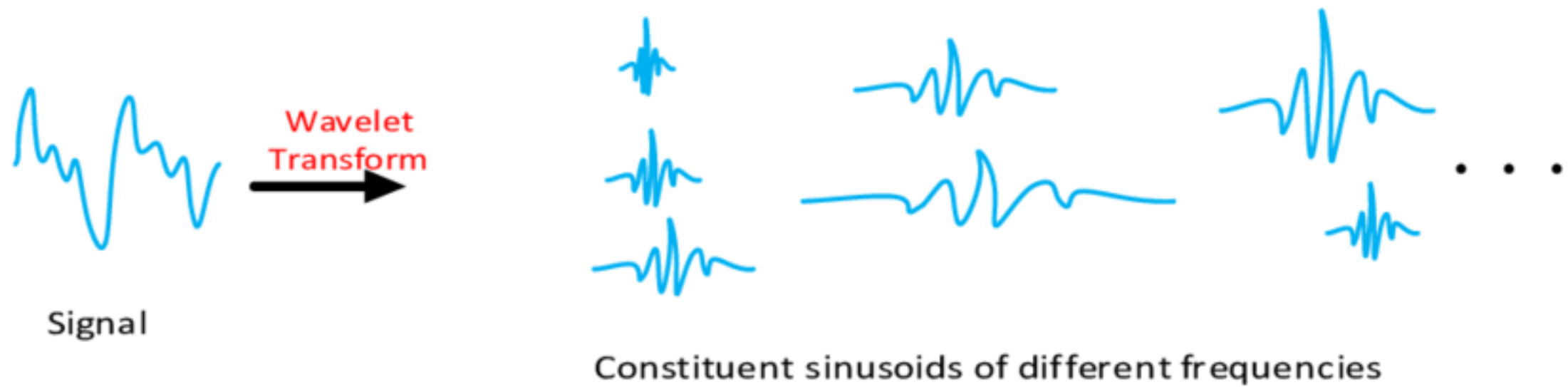
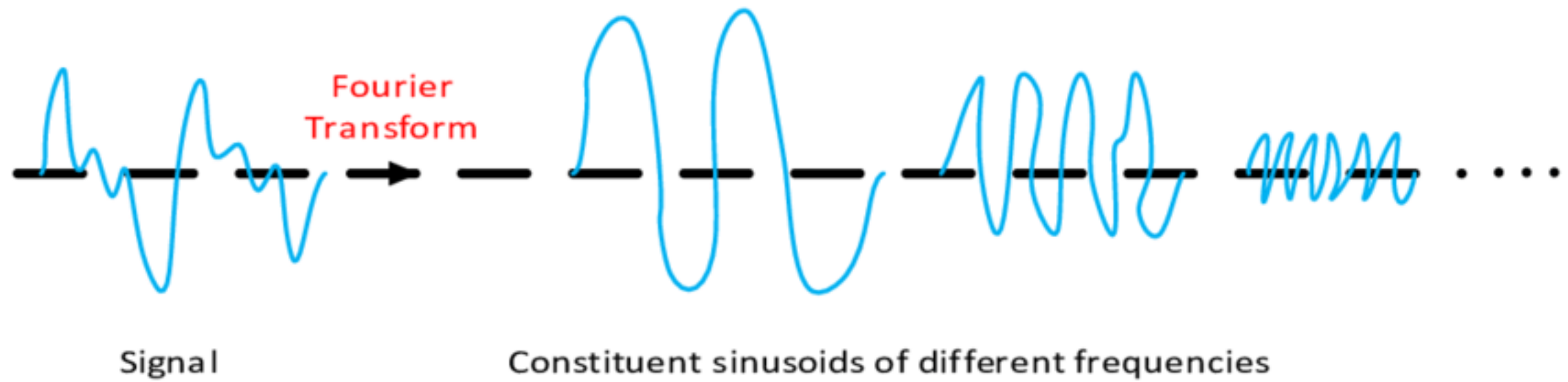
Wavelet Transform

The basis functions are small waves of varying frequency and short duration.

Converts signals into a series of wavelets by scaling and shifting the mother wavelet.

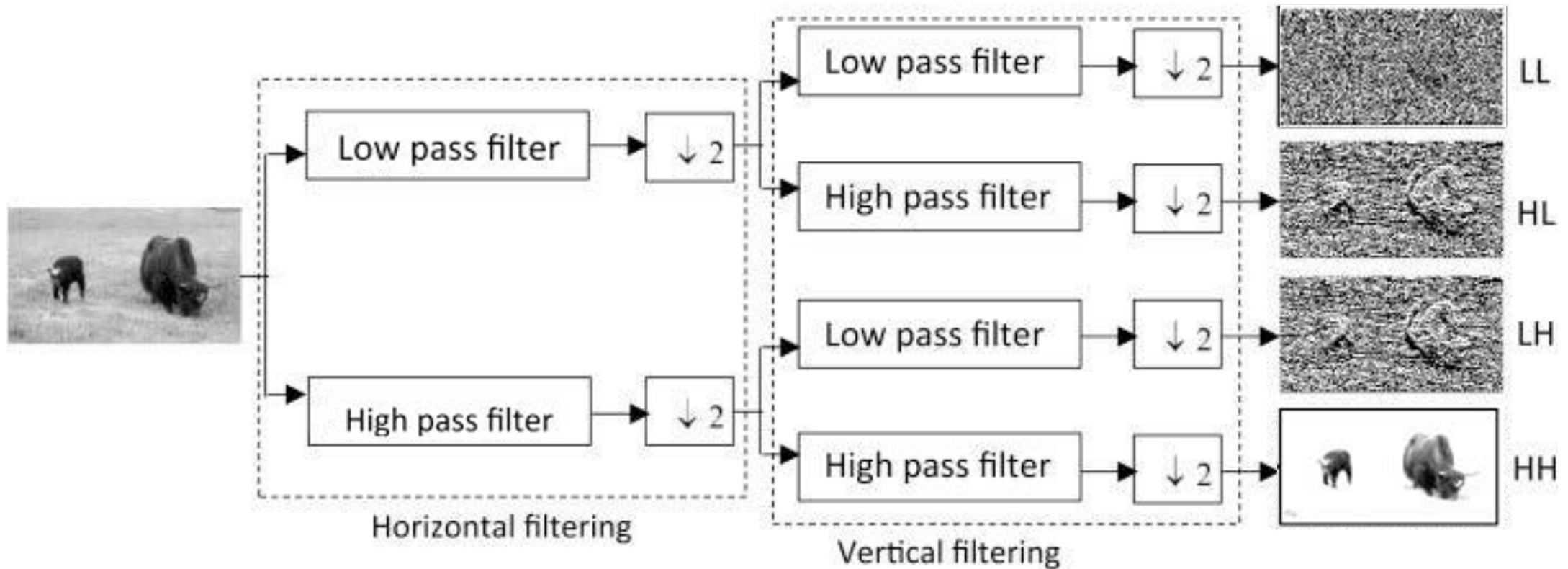
Provides both time and frequency information.

Temporal (time) information is preserved during transformation.

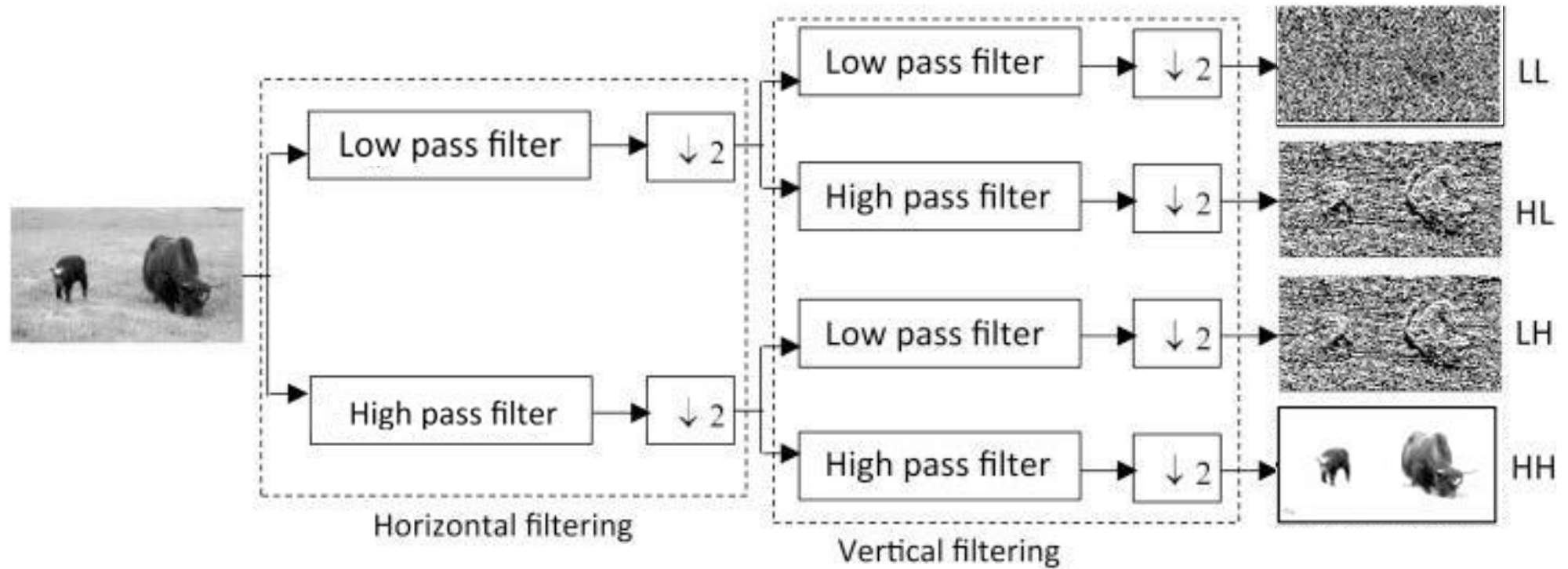


3. Wavelet Analysis in Image Processing

Wavelet analysis is used to divide information present on an image (signals) into two discrete components — approximations (low pass) and details (high pass).



A signal is passed through two filters, high pass and low pass filters. The image is then decomposed into high frequency (details) and low frequency components (approximation). At every level, we get 4 sub-signals. The approximation shows an overall trend of pixel values and the details as the horizontal, vertical and diagonal components.



If these details are insignificant, they can be valued as zero without significant impact on the image, thereby achieving filtering and compression.

3.1 Multiresolution Analysis (MRA)

Since wavelets can be scaled, WT allows analyzing signals at different levels of detail.

- ❑ Low frequencies (large scales) capture global trends or coarse details.
- ❑ High frequencies (small scales) capture fine details or edges.

This property makes wavelets highly useful in image processing, compression, noise reduction, and feature extraction.

3.1.1 How Humans Perceive Images

When we look at an image, we naturally group similar regions based on texture and intensity to identify objects.

- ❑ *Small objects or low-contrast details require a high-resolution view* for better visibility.
- ❑ *Large objects or high-contrast regions can be understood even with a low-resolution view.*

3.1.2 Why Use Multiresolution Analysis?

If an image contains both small and large objects, or low- and high-contrast regions, analyzing it at different resolutions helps capture all important details.

This is especially useful in image processing applications like:

- ✓ Image compression (JPEG 2000 uses wavelets)
- ✓ Feature extraction
- ✓ Denoising and edge detection
- ✓ Object recognition

Think of Google Maps:

- At a high zoom level, we see streets and small details (high resolution).
- At a low zoom level, we see entire cities and major roads (low resolution).

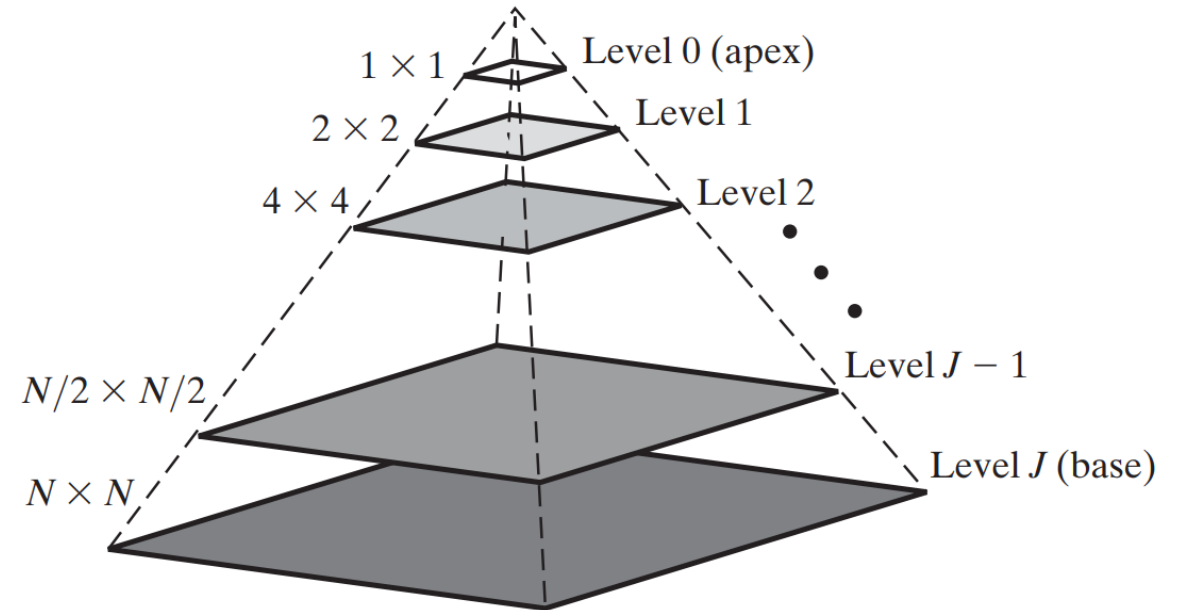
By combining multiple levels, we get a complete understanding of the area.

3.2 Image Pyramids

Originally devised for machine vision and image compression applications, an image pyramid is a collection of decreasing resolution images arranged in the shape of a pyramid.

The base of the pyramid contains a high-resolution representation of the image being processed; the apex contains a low-resolution approximation.

As we move up the pyramid, both size and resolution decrease.



3.2.1 Decomposition (Wavelet Transform Process)

At each level of the wavelet transform, an image is split into approximation (low-frequency) and detail (residual) components (high-frequency).

Instead of storing the full image at each level, we store only the difference (residuals) between the original image at that level and its approximation from the next level.

*** The approximation represents the main structure (low-frequency content), while the residuals contain finer details (edges, textures, noise).

Low-pass filter (scaling function): $h[n]$

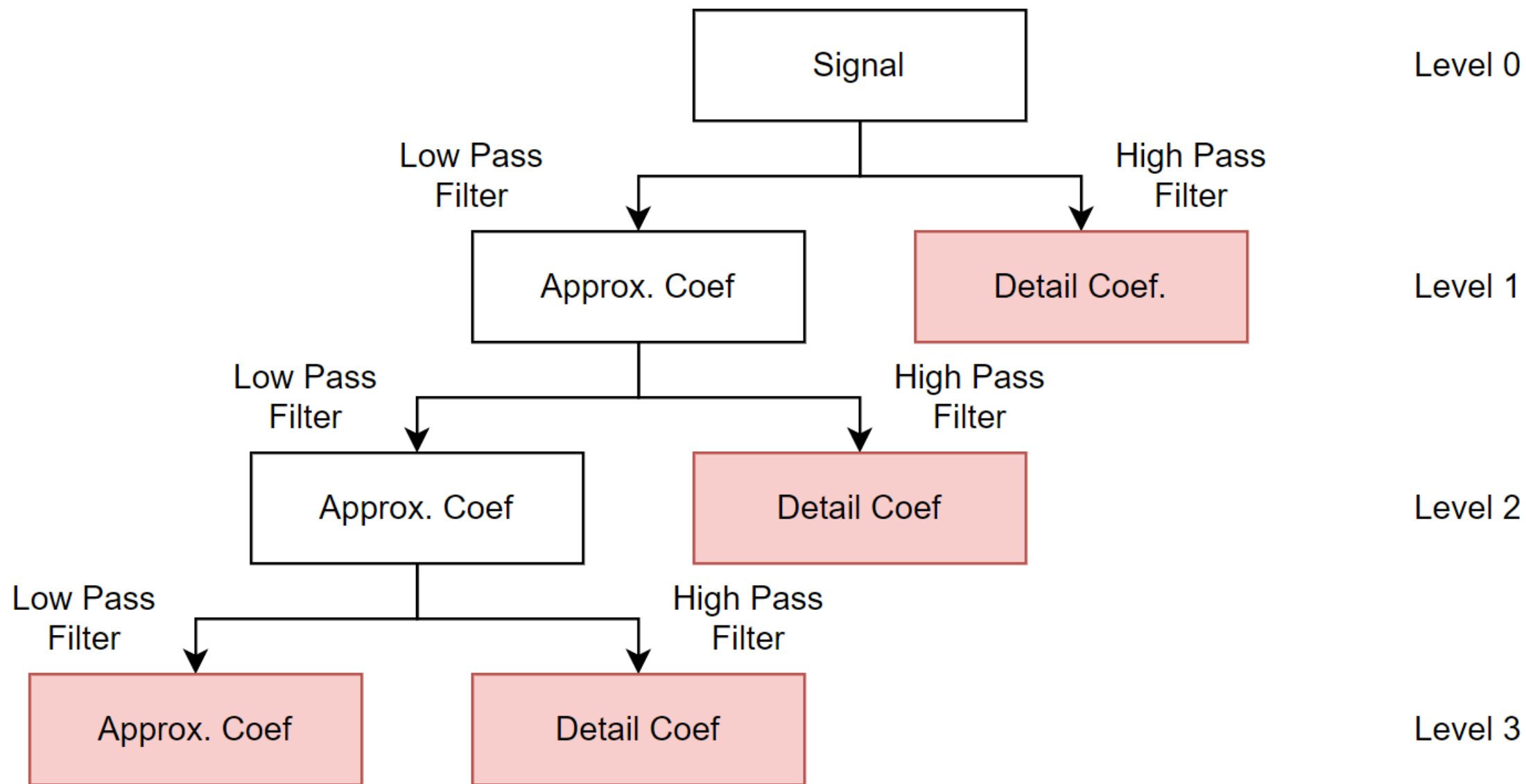
High-pass filter (wavelet function): $g[n]$

The approximation and detail coefficients are computed as:

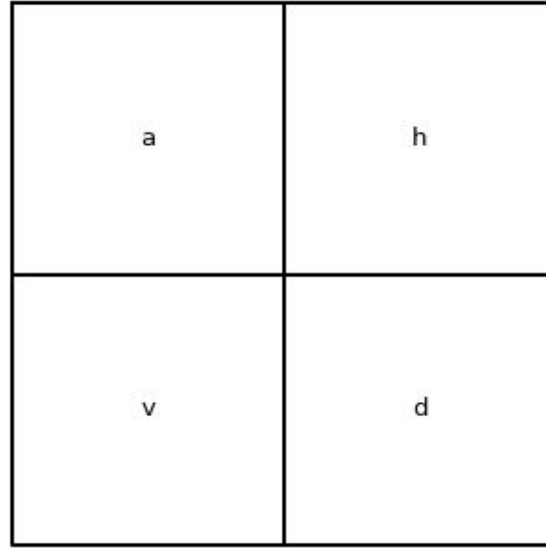
$$a[k] = \sum_n x[n] \cdot h[2k - n]$$

$$d[k] = \sum_n x[n] \cdot g[2k - n]$$

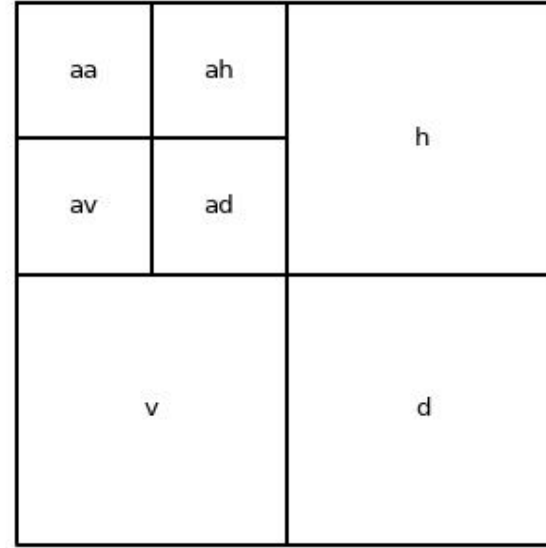
where $a[k]$ are the approximation coefficients, $d[k]$ are the detail coefficients, and k is the index of the transformed coefficients.



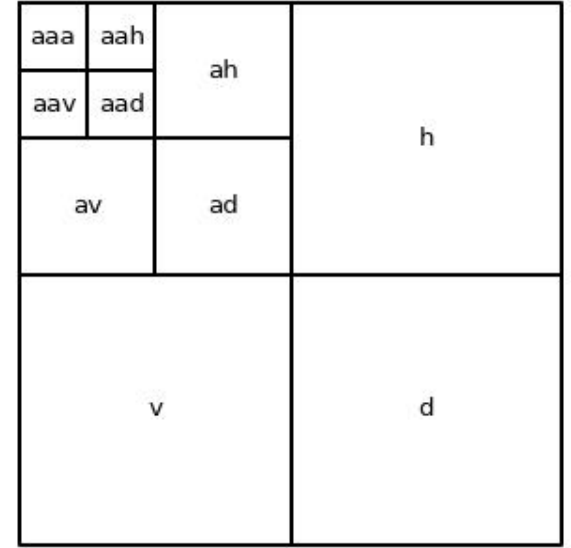
1 level
decomposition



2 level
decomposition



3 level
decomposition



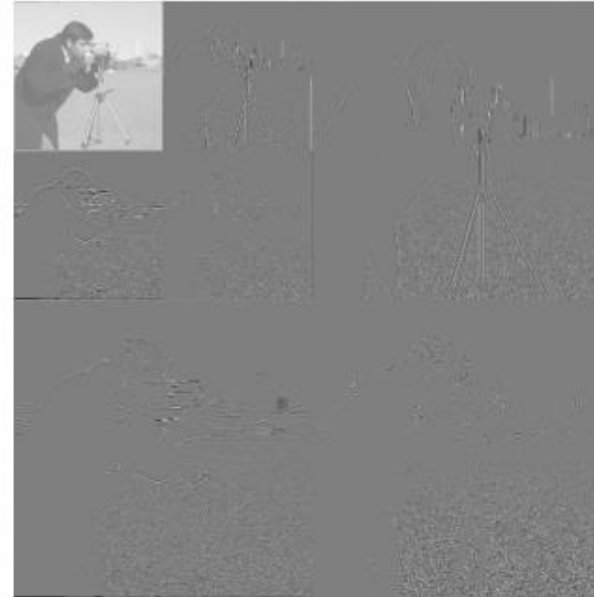
Image



Coefficients
(1 level)



Coefficients
(2 level)



Coefficients
(3 level)



3.2.2 Reconstruction (Inverse Wavelet Transform)

To reconstruct the image, we simply add up all the stored residuals, moving from the coarsest (lowest resolution) to the finest (highest resolution).

Since each level only stores differences, the full image can be perfectly rebuilt step by step.

